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AI Mobile Engineer

What if your AI could answer questions from your own documents instead of relying only on its training data?





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What is RAG?

- RAG (Retrieval-Augmented Generation): Retrieves relevant information before generating an AI response.
- Why RAG? Helps AI answer using your own documents and data.
- Example: Ask an AI study assistant a question about your uploaded PDF notes.





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Basic RAG Workflow

- Documents: Upload PDFs, notes, or text files.
- Chunking: Split large documents into smaller pieces.
- Embeddings: Convert text chunks into numerical vectors.
- Storage: Save embeddings and text in a Vector Database.
- Retrieval: Find chunks relevant to the user's question.
- Generation: Send retrieved information to the LLM to generate an answer.
- Sources: Return the answer with references to the original documents.





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What is Chunking?

- Splits large documents into smaller text sections.
- Makes information easier to search and retrieve.
- Example: Split a 100-page PDF into smaller chunks of 500 words.
- Overlap: Keeps some repeated text between chunks to preserve context.

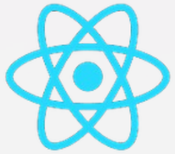




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What are Embeddings?

- Embeddings convert text into numerical vectors.
- Similar meanings produce vectors that are closer in vector space.
- Example: “How do I learn React Native?” and “React Native learning roadmap” may have similar embeddings.





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What is a Vector Database?

- Stores and searches embeddings efficiently.
- Finds information based on semantic similarity, not just exact keyword matching.
- Stores vectors alongside text and metadata.
- Retrieves relevant chunks when users ask questions.





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Popular Vector Database Options

- Supabase + pgvector: PostgreSQL database with vector storage and similarity search.
- ChromaDB: Developer-friendly vector database for AI and RAG applications.
- Pinecone: Managed vector database service designed for scalable similarity search.





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RAG in a Mobile App

- Mobile app sends the user's question to the AI backend.
- Backend generates query embeddings.
- Vector database retrieves relevant chunks.
- LLM generates an answer using retrieved context.
- Backend returns the answer and sources to the mobile app.



